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Blood-based circulating tumour DNA (ctDNA) tests for colorectal cancer screening: Systematic review and meta-analysis of diagnostic accuracy

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ABSTRACT

Background: Blood-based circulating tumour DNA (ctDNA) assays have emerged as a promising tool for minimally invasive colorectal cancer (CRC) screening. However, their diagnostic accuracy in asymptomatic, average-risk populations remains uncertain. This systematic review and meta-analysis aimed to synthesise current evidence on the performance of ctDNA-based blood tests for detecting advanced colorectal neoplasia (ACN), defined as the composite of invasive CRC and advanced precancerous lesions (APL).

Methods: A comprehensive search of PubMed, EMBASE, and the Cochrane Library was conducted through July 2025. Studies evaluating ctDNA-based blood assays against colonoscopy and histopathology in asymptomatic, average-risk adults were included. Pooled estimates were calculated using a bivariate random-effects model following Cochrane DTA guidance.

Results: Three population-based prospective studies were included (n = 36,381). For invasive CRC, pooled sensitivity was 0.72 (95% CI 0.49–0.88) and specificity 0.91 (95% CI 0.89–0.92), with an area under the curve (AUC) of 0.92. Sensitivity increased progressively from stage I (0.53) to stage IV (0.89). For APL, pooled sensitivity was 0.13 (95% CI 0.12–0.14) and specificity 0.90 (95% CI 0.88–0.91). When CRC and APL were considered together (ACN), overall sensitivity was 0.16 (95% CI 0.14–0.18) with specificity 0.91 (95% CI 0.89–0.92).

Conclusions: ctDNA-based blood testing demonstrates high specificity and clinically relevant accuracy for invasive CRC, but limited detection of precancerous lesions. These findings consolidate current evidence by defining the complementary role of ctDNA in population screening. As a non-invasive, patient-centred approach, ctDNA testing could increase participation and access in CRC prevention, but large-scale studies are needed to confirm its clinical and economic viability.

1. Introduction

Colorectal cancer (CRC) remains among the leading causes of cancer-related mortality worldwide, with over 1.9 million new cases and nearly one million deaths each year (Siegel et al., 2024). The progressive shift towards younger age at diagnosis, together with the observation that early-onset CRCs are often detected at more advanced stages than late-onset cases, highlights the importance of starting screening at earlier ages. (Carbone et al., 2025a) Although population screening has demonstrably reduced both CRC incidence and mortality through the identification of pre-cancerous lesions and early diagnosis of CRC, (Mandel et al., 1993; Kronborg et al., 1996) global adherence to recommended programmes remains suboptimal, mainly due to the invasiveness, preparation requirements, and perceived discomfort associated

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with colonoscopy and stool-based tests. (Ebner et al., 2024; Bridou et al., 2013; Paudel et al., 2025) Consequently, there is increasing interest in developing non-invasive, scalable, and patient-friendly approaches for early CRC detection that could enhance screening participation and coverage. (Shaukat and Levin, 2022)

Circulating tumour DNA (ctDNA) is a tumour-derived component of cell-free DNA (cfDNA) released into the bloodstream through apoptosis and necrosis of cancer cells. Advances in next-generation sequencing (NGS) and methylation-based assays have enabled highly sensitive detection of these tumour signatures, leading to the development of blood-based tests aimed at detecting colorectal neoplasia at its earliest stages. (Ma et al., 2024) Beyond ctDNA methylation assays, a broad range of molecular biomarkers, including circulating microRNA polymorphisms and other molecular drivers of colorectal carcinogenesis, have been investigated to improve risk stratification and early detection. (Yu et al., 2026; Barahman et al., 2024)

Although emerging data support the feasibility of ctDNA-based testing for detecting invasive CRC with reasonable accuracy, the reported sensitivity for advanced precancerous lesions (APL) remains low, and its performance in asymptomatic, average-risk populations in real-world settings is still uncertain. (Mannucci and Goel, 2025; Shaukat and Levin, 2022; Bessa et al., 2023)

The present systematic review and meta-analysis aim to synthesise current evidence on the diagnostic accuracy of blood-based ctDNA testing for detecting advanced colorectal neoplasia (ACN), CRC, and APL in asymptomatic, average-risk individuals. By providing pooled estimates of sensitivity, specificity, and overall discriminative performance, this study seeks to clarify the potential role of ctDNA-based assays within the evolving landscape of precision CRC prevention and population-wide screening.

2. Methods

2.1. Aim of the study and index test

The study aims to evaluate the performance of ctDNA for detecting ACN, CRC, and APL in an asymptomatic population relevant to screening, through a systematic synthesis and meta-analysis of the current evidence. In particular, the study aims to describe the sensitivity, specificity, positive and negative likelihood ratios (LR+, LR-) of tumour-specific ctDNA in diagnosing sporadic CRC and APL. This study was conducted using the Cochrane methodology for Diagnostic Test Accuracy (DTA) (Deeks et al., 2023) and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) checklist. (Page MJ et al., 2021) The protocol was registered in PROSPERO with the number CRD420251163485. (Carbone et al., 2025b)

2.2. Target population

The target population consisted of asymptomatic individuals at average risk for CRC and over the age of 40. Individuals with a personal history of inflammatory bowel disease or a family history of hereditary colorectal syndrome were excluded.

2.3. Target conditions and reference standard

The primary outcomes of interest were the sensitivity, specificity, and accuracy performance of ctDNA-based blood tests for detecting ACN and CRC. ACN was defined as the set of diagnoses that includes colorectal adenocarcinoma (CRC) and APL, the latter defined as adenomas ≥ 10 mm in size, adenomas with villous architecture, serrated lesions ≥ 10 mm in size, high-grade dysplasia, or intramucosal cancer.

Secondary outcomes included the ctDNA test accuracy for detecting APL alone and the sensitivity stratified by CRC stage.

Colonoscopy with histopathological assessment was considered the reference standard for all outcomes. When multiple lesions were found, the most clinically significant one was defined as the index lesion. The colonoscopy could have been performed before, at the same time, or after the ctDNA test, but no more than 4 months later.

2.4. Selection criteria for studies

All studies reporting the primary DTA estimates (true positives and negatives, false positives and negatives) of ctDNA for detecting sporadic CRC and/or ACN, compared with a definitive diagnosis made by standard colonoscopy and histopathology, in an average-risk patient population (target population), were assessed for inclusion. Only original studies with published full articles were included. Reviews, abstracts, conference papers, letters, and editorials were excluded. Only studies published in English were considered. Only studies with a single target population, a description of the index test (tumour-specific ctDNA), and the use of the reference standard (colonoscopy with histopathology) were included. Studies with a two- or multi-group design (proof-of-concept studies) were excluded. For this review, ctDNA-based tests were defined as blood-based assays detecting tumour-specific features (e.g., methylation, mutations, or tumour-informed fragmentomics). Studies assessing only total cfDNA quantity or non-specific fragment length distributions were excluded.

2.5. Search, screening and data extraction

The literature search was conducted across the EMBASE-Medline, PubMed, and Cochrane Library databases without time restrictions until July 5, 2025. The search query included main words such as "circulating-tumour DNA (ctDNA)", "cell-free DNA (cfDNA)", "colorectal cancer", "screening", and “colonoscopy” (the complete search query is reported in Appendix A in the Supplementary Materials). After removing duplicates, two authors (FC and DC) examined all the studies based on their titles and abstracts using the Rayyan web application. (Ouzzani et al., 2016) The studies identified were screened against the inclusion criteria in full text. A hand-search of the references from the included studies and previous reviews on the topic was also conducted to identify additional relevant studies, adhering to established selection criteria. Any incongruences between the two reviewers were resolved through discussion with a third researcher (SG).

The outcomes extracted from the studies were population characteristics, ctDNA extraction techniques, the number of CRCs detected by colonoscopies, the CRC stage at diagnosis, the number of precancerous lesions detected by colonoscopy, and the number of true and false positive and negative events between ctDNA and colonoscopy for CRC and precancerous lesions.

2.6. Risk-of-bias appraisal and certainty of the evidence

The methodological quality of included diagnostic accuracy studies was assessed using the Quality Assessment of Diagnostic Accuracy Studies−2 (QUADAS−2) tool, in accordance with Cochrane recommendations for DTA reviews. (Whiting et al., 2011) Each study was categorised as having a “low,” “high,” or “unclear” risk of bias. Publication bias was planned to be assessed if more than 5 studies were included.

The overall certainty of the evidence was assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach adapted for DTA studies, considering the risk of bias, indirectness, inconsistency, imprecision, and publication bias. (Schünemann et al., 2020a, b) Certainty of the evidence for each outcome was rated as high, moderate, low, or very low.

2.7. Statistics

Diagnostic accuracy was assessed for the three predefined endpoints:

ACN, CRC and APL. For each endpoint, outcome measures included pooled estimates of sensitivity (Sn), specificity (Sp), positive likelihood ratio (LR+), negative likelihood ratio (LR-), diagnostic odds ratio (DOR) and their 95% confidence intervals (CI). Individuals without the target condition were treated as the negative reference group for each endpoint. Summary receiver operating characteristics (SROC) curves with their relative 95% confidence contour and 95% prediction contour were produced, and the area under the curve (AUC) was calculated as a global measure of performance for each diagnostic test. For this purpose, bivariate models based on restricted maximum likelihood (REML) were developed.

Test accuracy-specific Zhou and Dendukuri I^2 statistics were computed, (Zhou and Dendukuri, 2014) and a prediction region around the summary point estimate was included in the SROC plot to display the magnitude of the observed heterogeneity graphically. (Deeks et al., 2023)

Statistical analysis was conducted with STATA 17 [StataCorp. 2021. *Stata Statistical Software: Release 17*. College Station, TX: StataCorp LLC] using the *metaDTA* package, (Dwamena, 2009) and R statistical software (The Comprehensive R Archive Network), with *mada*, *lme4*, and *robvis* packages. (Bates et al., 2015; McGuinness and Higgins, 2021; Doebler and Holling, 2015)

3. Results

3.1. Descriptive analysis of included studies

A total of 227 studies were identified through database searching; after removing 47 duplicates, 180 documents were screened based on title and abstract, of which 33 were assessed for full text. Two studies were identified through citation searching and subsequently evaluated for full text. Three studies (Shaukat et al., 2025; Chung DC et al., 2024; Church TR et al., 2014) met the selection criteria and were included in both the qualitative and quantitative meta-analyses. The PRISMA flow diagram is shown in Fig. 1.

The studies reported a total of 41,745 patients, with a 10% prevalence for ACN (n = 4539), 0.5% for CRC (n = 190), and 10% for APL (n = 4349). In the PRESEPT study by Church et al. (Church TR et al., 2014), only a subset of the enrolled screening population (n = 1510) was included in the per-protocol diagnostic accuracy analysis. Therefore, the total number of patients included in the meta-analysis was 36,381, comprising 4187 ACN (12%), 190 CRC (0.5%), and 3997 APL (11%). Characteristics of included studies are reported in Table 1.

The PRESEPT study (Church TR et al., 2014) used weighted estimates of Sn and Sp that accounted for stratified sampling (all colorectal cancers and a random non-CRC subsample). The actual cell counts used in this analysis reflect these adjustments, not the raw laboratory data.

3.2. Diagnostic performance for ACN, CRC and APL

The pooled results for the identification of ACN were Sn = 0.16 (95% CI: 0.14–0.18), Sp = 0.91 (95% CI: 0.89–0.92), LR + = 1.69 (95% CI: 1.56–1.83), LR- = 0.93 (95% CI: 0.92–0.94), DOR = 1.81 (95% CI: 1.65–1.99), and AUC = 0.55 (95% CI: 0.39–0.70). No heterogeneity was detected (I^2 = 0%).

The pooled results for the detection of CRC were Sn = 0.72 (95% CI: 0.49–0.88), Sp = 0.91 (95% CI: 0.89–0.92), LR + = 7.70 (95% CI: 6.40–9.28), LR- = 0.30 (95% CI: 0.14–0.62), DOR = 25.21 (95% CI: 10.26–61.94) and AUC = 0.92 (95% CI: 0.90–0.98). High heterogeneity was detected (I^2 = 80.2%).

To explore assay-related heterogeneity, a sensitivity analysis excluding the PCR-based mSEPT9 assay (PRESEPT), representing an earlier-generation platform, was performed. When restricted to the two multitarget methylation NGS studies, pooled CRC sensitivity increased

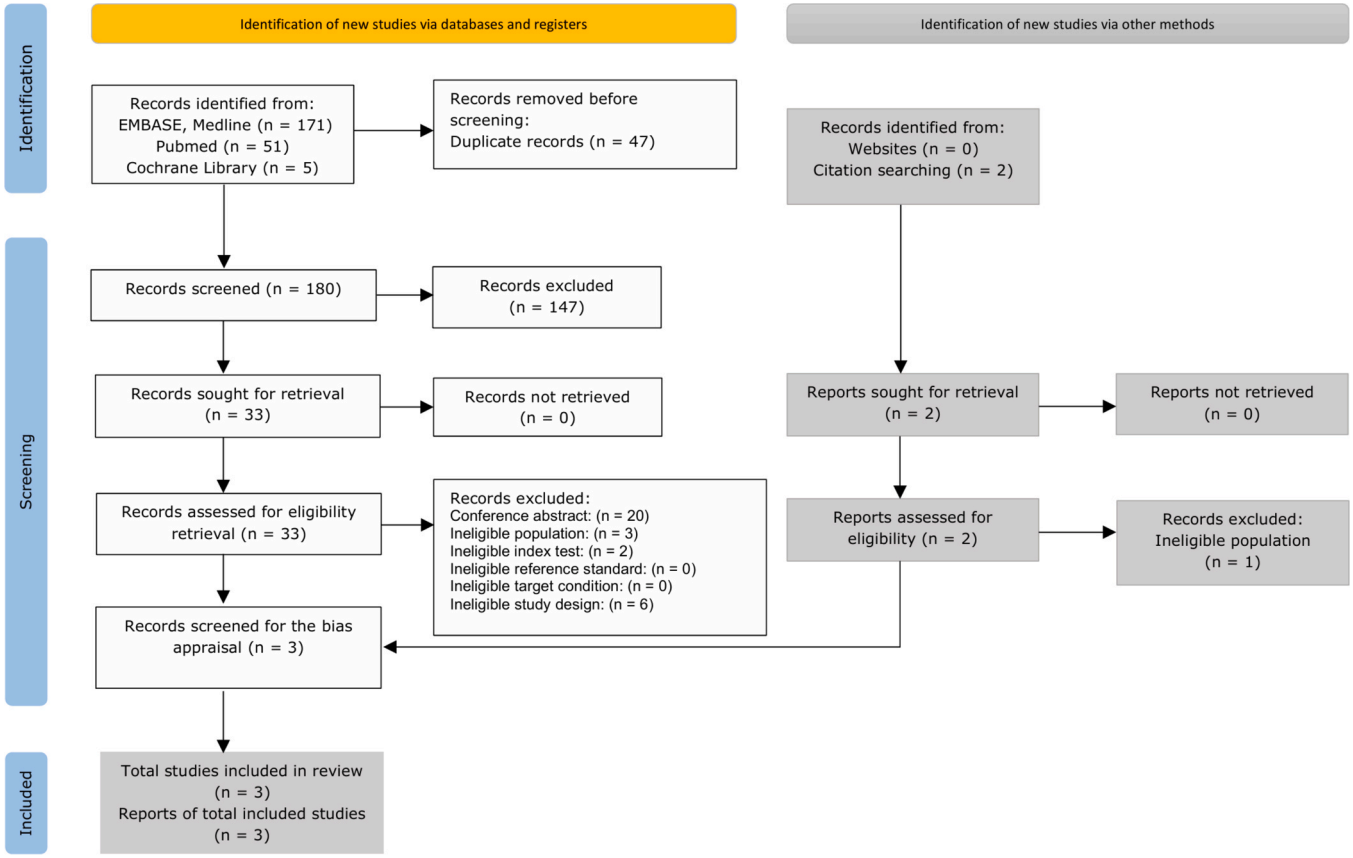


Fig. 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram.

Table 1
Characteristics of included studies.

Study	Country	Period of inclusion	ctDNA extraction technique	Age (range)	N. of patients [n = 41,745]	Gender, male (%): female (%) [18,681 (45): 23,064 (55)]	ACN, n (%) [4539 (11)]	CRC, n (%) [190 (0.5)]	APL, n (%) [4349 (10)]
Shaukat et al. 2025 (PREEMPT)	US, United Arab Emirates	2020–2022	Plasma cfDNA targeted methylation sequencing assay (multitarget cfDNA methylation, Guardant SHIELD® test)	45–85	27010	11934 (44) : 15076 (56)	2639 (9.7)	72 (0.3)	2567 (9.5)
Chung et al. 2024 (ECLIPSE)	US	2019–2022	Plasma cfDNA targeted methylation assay with next-generation sequencing (multitarget methylation NGS panel; Guardant SHIELD® test)	45–84	7861	3643 (46) : 4218 (54)	1181 (15.0)	65 (0.8)	1116 (14.2)
Church et al. 2014 (PRESEPT)	US, Germany	2008–2010	Plasma mSEPT9 methylation assay (Epi proColon® 2.0; real-time PCR-based)	50–80	6874*	3104 (45) : 3770 (55)*	719 (10.5)*	53 (0.8)	666 (9.7)*

ctDNA: circulating tumour DNA; N.: number; ACN: advanced colorectal neoplasia; CRC: colorectal cancer; APL: advanced precancerous lesion; cfDNA: circulating free DNA; NGS: next generation sequencing.
* In Church et al. 2014, only a sample of the non-CRC group was analysed per protocol and was meta-analysed: 1510 patients; gender: M 689 (45.6%), F 821 (54.4%).

to 0.81 (95% CI 0.73–0.88), specificity was 0.90 (95% CI 0.88–0.92), and the AUC remained high at 0.93, with no residual between-study heterogeneity observed ($I^2 = 0\%$).

The pooled results for the detection of APL were Sn = 0.13 (95% CI: 0.12–0.14), Sp = 0.90 (95% CI: 0.88–0.91), LR + = 1.27 (95% CI: 1.07–1.52), LR- = 0.97 (95% CI: 0.95–0.99), DOR = 1.31 (95% CI: 1.08–1.59), and AUC = 0.17 (95% CI: 0.07–0.26). Very low heterogeneity was detected ($I^2 = 0.3\%$).

Given the small number of included studies, AUC estimates, particularly for APL, may be unstable and should be interpreted cautiously. For screening interpretation, it is preferable to refer to pooled sensitivity and specificity.

Forest plots of Sn and Sp are shown in Fig. 2. The SROC curves are shown in Fig. 3.

3.3. Expected screening outcomes

To facilitate interpretation in a screening setting, pooled CRC sensitivity (0.72) and specificity (0.91) were applied to the meta-analytic CRC prevalence of 0.5%. In a hypothetical cohort of 10,000

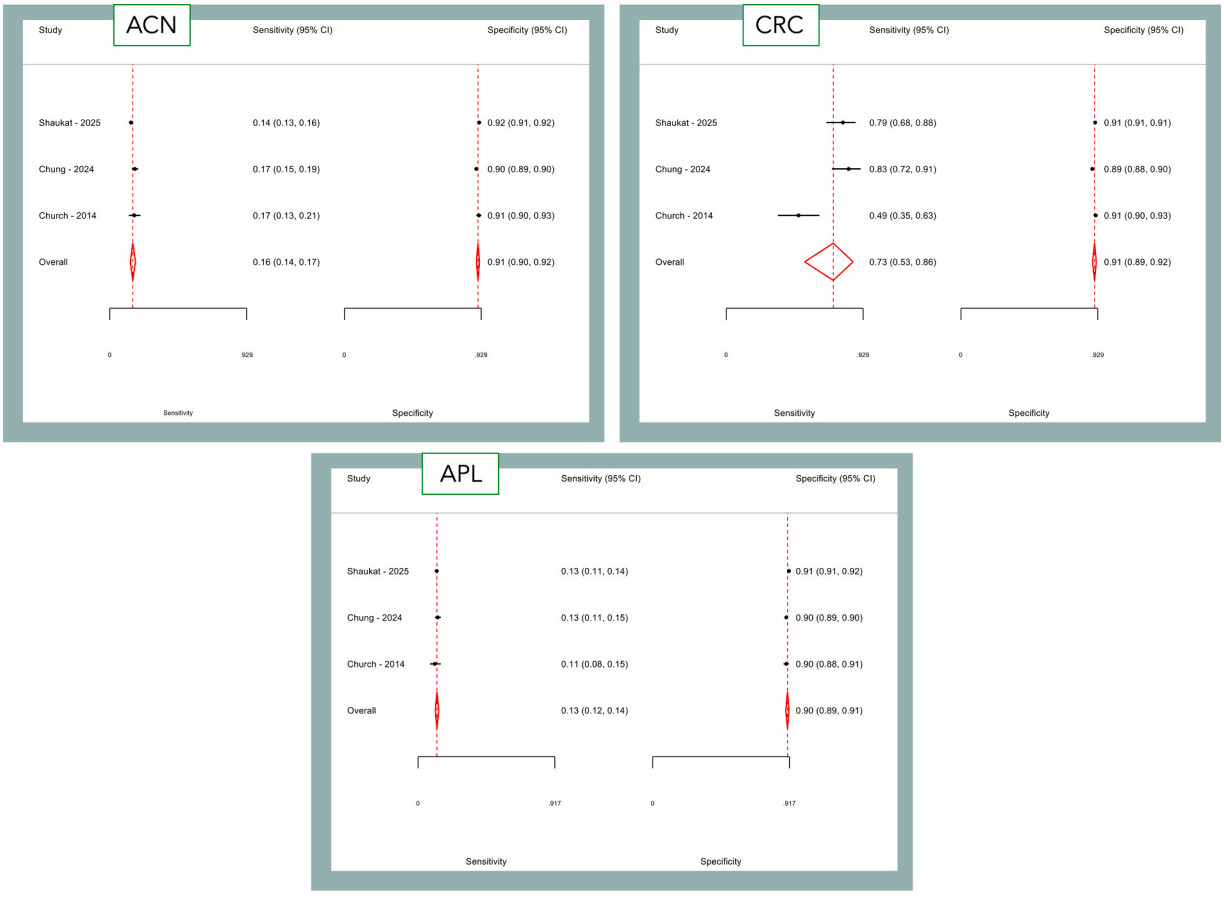


Fig. 2. Forest plots of sensitivity and specificity of circulating tumour DNA (ctDNA) in detecting advanced colorectal neoplasia (ACN), colorectal cancer (CRC), and advanced precancerous lesions (APL).

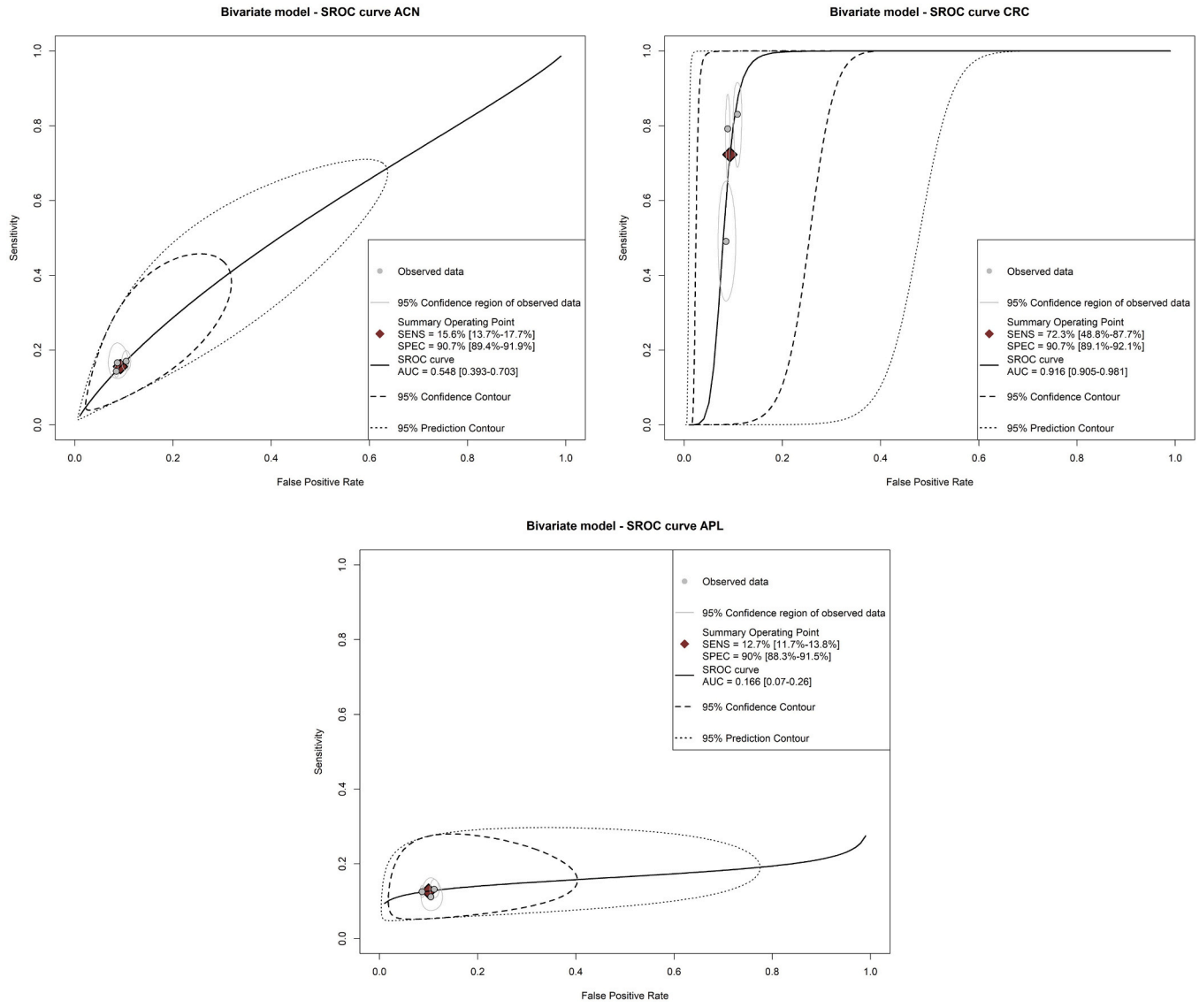


Fig. 3. Summary receiver operating characteristic (SROC) curves for advanced colorectal neoplasia (ACN), colorectal cancer (CRC), and advanced precancerous lesions (APL).

screened individuals, approximately 50 CRC cases would be expected, of which 36 would be detected (true positives, 0.4%), and 14 would be missed (false negatives, 0.1%). Among 9950 individuals without CRC, around 896 would test positive (false positives, 9%) and 9054 would test negative (true negatives, 90.5%). Under these assumptions, the estimated positive predictive value (PPV) would be approximately 3.9%, while the negative predictive value (NPV) would be approximately 99.8% (Appendix B, Table B.1, Supplementary Materials).

3.4. Diagnostic performance for CRC stages

The pooled results for CRC stage I disease were Sn = 0.53 (95% CI: 0.37–0.69), Sp = 0.90 (95% CI: 0.89–0.92), LR + = 5.61 (95% CI: 4.40–7.15), LR- = 0.55 (95% CI: 0.39–0.77), DOR = 10.22 (95% CI: 5.55–18.79), and AUC = 0.89 (95% CI: 0.82–0.96). Heterogeneity was high (I^2 = 77.8%).

The pooled results for CRC stage II disease were Sn = 0.89 (95% CI: 0.49–0.98), Sp = 0.90 (95% CI: 0.89–0.92), LR + = 9.07 (95% CI: 7.31–11.25), LR- = 0.12 (95% CI: 0.02–0.80), DOR = 75.86 (95% CI: 8.77–655.83), and AUC = 0.92 (95% CI: 0.86–0.98). Heterogeneity was high (I^2 = 86.6%).

The pooled results for CRC stage III disease were Sn = 0.82 (95% CI: 0.38–0.97), Sp = 0.90 (95% CI: 0.89–0.92), LR + = 8.42 (95% CI: 7.16–9.90), LR- = 0.26 (95% CI: 0.09–0.81), DOR = 32.74 (95% CI: 6.83–156.91), and AUC = 0.91 (95% CI: 0.85–0.97). Heterogeneity was high (I^2 = 84.2%).

The pooled results for CRC stage IV disease were Sn = 0.89 (95% CI: 0.64–0.97), Sp = 0.90 (95% CI: 0.89–0.91), LR + = 9.25 (95% CI: 7.58–11.30), LR- = 0.15 (95% CI: 0.05–0.48), DOR = 68.47 (95% CI: 17.54–267.24), and AUC = 0.93 (95% CI: 0.87–0.99). Heterogeneity was high (I^2 = 83.7%).

SROC curves for the CRC stages are reported in Appendix C of the Supplementary Materials.

3.5. Risk of bias, publication bias and certainty of evidence assessment

According to the QUADAS−2 risk-of-bias assessment, two studies (Shaukat et al., 2025; Chung et al., 2024) were rated "unclear" in the *flow and timing* domain, and one study (Church TR et al., 2014) was rated "unclear" in the *patient selection* domain. Although the overall risk of bias was rated unclear for all studies, the overall applicability concerns were predominantly rated as "low."

The risk-of-bias assessment plots are reported in Appendix D in the Supplementary Materials.

The number of included articles was too small to assess publication bias.

The certainty of evidence was low for ACN, moderate for CRC and very low for APL detection. The GRADE summary of findings is reported in Table 2.

4. Discussion

Blood-based assays using ctDNA have emerged as a promising advance in CRC screening. In this meta-analysis, ctDNA testing showed relatively high specificity (91%) and moderate sensitivity (72%) for invasive CRC. However, when projected onto a screening population with low disease prevalence, the resulting PPV remains low, with a substantial number of false positive results expected. The diagnostic yield for APL remained consistently low (Sn 13%, Sp 90%), indicating that current ctDNA assays are more effective at detecting established malignancies than early neoplastic changes. (Shaukat et al., 2025; Chung et al., 2024; Bessa et al., 2023;)

Because APL prevalence (~10%) substantially exceeds CRC prevalence (~0.5%) in screening populations, pooled ACN sensitivity is largely driven by APL detection. The low ACN sensitivity observed (16%), therefore, primarily reflects limited identification of APLs rather than poor detection of invasive CRC.

This diagnostic imbalance highlights the biological and technical hurdles in detecting minimal residual tumour DNA in asymptomatic, average-risk individuals, where the tumour-derived cfDNA fraction is exceedingly limited. (Tsui et al., 2025)

Overall, these findings define the current boundary between technological capability and biological detectability, positioning ctDNA testing as a promising yet still developing element within the landscape of precision CRC prevention.

Compared with established screening methods, the overall accuracy of ctDNA testing is similar to that of faecal immunochemical testing (FIT), which has a pooled sensitivity for CRC of 71–78% and specificity of 90–95%, while its pooled sensitivity for advanced adenoma is 27–31%. (Imperiale et al., 2019) When translated into expected outcomes per 10,000 screened individuals, ctDNA testing would generate approximately 896 false positive results (9%) compared with 497 (5%) using FIT at a 20 μg/g threshold, despite comparable sensitivity for invasive CRC (Appendix B in the Supplementary Materials). This difference in downstream colonoscopy burden is clinically relevant and should be considered when evaluating implementation strategies.

Stool-based DNA or RNA tests demonstrate higher sensitivity for advanced adenomas (46%) at the expense of lower specificity (90%). (Ebner et al., 2025; Barnell et al., 2023; Imperiale et al., 2024)

From a screening perspective, detection of APL and the resulting colonoscopy burden are critical determinants of clinical impact, and in this regard, ctDNA assays currently demonstrate limited sensitivity for

Table 2
Summary of findings and certainty of the evidence (GRADE) for ctDNA-based blood testing in colorectal cancer screening.

Outcome	No. of studies (events/ participants)	Effect estimates (95% CI)	Certainty of the evidence (GRADE)	Reasons for rating
Advanced colorectal neoplasia (ACN)	3 studies (4187/36,381)	Sn = 0.16 (0.14–0.18) Sp = 0.91 (0.89–0.92) LR + = 1.69 (1.56–1.83) LR- = 0.93 (0.92–0.94) DOR = 1.81 (1.65–1.99) AUC = 0.55 (0.39–0.70)	⊕⊕⊕⊖ Moderate	• Risk of bias ↓ – minor concerns in patient selection and flow/timing • Indirectness – no • Inconsistency – no (I^2 = 0%) • Imprecision – no, few true positives (1.8%) but narrow 95%CIs
Colorectal cancer (CRC)	3 studies (190/36,381)	Sn = 0.72 (0.49–0.88) Sp = 0.91 (0.89–0.92) LR + = 7.70 (6.40–9.28) LR- = 0.30 (0.14–0.62) DOR = 25.21 (10.26–61.94) AUC = 0.92 (0.90–0.98)	⊕⊕⊕⊖ Moderate	• Risk of bias ↓ – minor concerns in patient selection and flow/timing • Indirectness – no • Inconsistency ↓ – high heterogeneity (I^2 ≈ 80%) explained by different assay designs and analytical thresholds • Imprecision – no, few true positives (0.4%) but narrow 95%CIs • Gradient ↑ – progressive increase in sensitivity across CRC stages (0.53→0.89) supports biological plausibility and strengthens confidence in the estimate
Advanced precancerous lesions (APL)	3 studies (3997/36,381)	Sn = 0.13 (0.12–0.14) Sp = 0.90 (0.88–0.91) LR + = 1.27 (1.07–1.52) LR- = 0.97 (0.95–0.99) DOR = 1.31 (1.08–1.59) AUC = 0.17 (0.07–0.26)	⊕⊕⊖⊖ Low	• Risk of bias ↓ – minor concerns in patient selection and flow/timing, minor variability in APL definitions across the studies • Indirectness ↓ – heterogeneous analytical platforms with different detection thresholds for non-invasive lesions may affect cross-study comparability • Inconsistency – no (I^2 ≈ 0%) • Imprecision – no, few true positives (1.4%) but narrow 95%CIs

GRADE: Grading of Recommendations Assessment, Development and Evaluation; Sn: sensitivity; Sp: specificity; LR + : positive likelihood ratio; LR-: negative likelihood ratio; DOR: diagnostic odds ratio; AUC: area under the curve.
AUC estimates for APL are derived from SROC modelling and may be unstable with few studies; interpretation should prioritise pooled sensitivity/specificity and likelihood ratios.
Publication bias was not assessable for all the outcomes.

APL and only moderate specificity.

The present analysis highlights a key clinical distinction: ctDNA testing achieves levels of specificity comparable to FIT, but its ability to detect preneoplastic lesions remains limited. The increasing sensitivity across tumour stages, from 53% in stage I to 89% in stage IV, further supports the biological dependence of ctDNA detectability on tumour burden and vascular invasion. This gradient suggests that ctDNA assays are currently more effective at identifying established malignancies rather than early-stage transformations. From a public health perspective, although blood-based testing may improve acceptability and participation, its current performance profile suggests that any potential gains in uptake must be balanced against the increased number of false positive results and associated colonoscopies in low-prevalence screening settings. (Ma et al., 2024; Bridou et al., 2013)

The low sensitivity for APLs likely reflects fundamental biological constraints. ctDNA release is driven by tumour cell turnover, necrosis, and vascular invasion, all of which are uncommon in non-invasive adenomas or early intramucosal lesions. Consequently, the circulating tumour fraction in plasma remains below the detection threshold of current assays. Furthermore, variations in methylation density, cfDNA fragment size, and background noise from non-tumour cell death may further dilute the tumour-specific signal and contribute to false-negative results. Improvements in sequencing depth, fragmentomic analysis, and machine-learning-based signal enrichment could potentially enhance early lesion detection, but these refinements remain at an exploratory stage. (Ma et al., 2024; Tsui et al., 2025)

From a clinical perspective, the findings of this meta-analysis support a potentially complementary rather than substitutive role for ctDNA testing within CRC screening programmes. High specificity and patient acceptability make blood-based testing an attractive first-line option for individuals unwilling or unable to undergo stool or endoscopic procedures. When used as an adjunctive tool, ctDNA assays could increase overall participation and extend screening reach to populations historically underrepresented in organised programmes, particularly younger adults and those experiencing screening fatigue. (Symonds et al., 2025; Coronado et al., 2024)

In this scenario, integration into existing algorithms should, however, be guided by a clear understanding of the test's current limitations. Given the low sensitivity for APLs, ctDNA screening alone cannot yet replace established stool-based or endoscopic modalities for cancer prevention. Instead, a pragmatic framework may involve sequential or risk-stratified approaches, using ctDNA to triage participants for colonoscopy or to maintain engagement between conventional screening rounds. Such hybrid models are already being explored in population-based studies and could ultimately redefine adherence and cost-effectiveness metrics within precision prevention strategies, although current economic evaluations indicate that neither stand-alone nor combined ctDNA-based approaches are yet cost-effective for population-wide screening. (Cairns et al., 2022; Ladabaum et al., 2020; Aziz et al., 2023)

This meta-analysis provides an up-to-date screening-focused synthesis of evidence on the DTA of blood-based ctDNA testing in CRC screening. By restricting inclusion to large, prospective, population-based studies conducted in asymptomatic, average-risk individuals, the present work provides robust pooled estimates derived from studies at lower risk of bias that closely reflect real-world screening performance.

Nevertheless, several limitations must be acknowledged. The review was restricted to English-language publications, which may introduce potential language bias. The small number of eligible studies limited the exploration of between-study heterogeneity and precluded a formal assessment of publication bias. Differences in assay design, ranging from single-gene methylation assays (mSEPT9) (Church et al., 2014) to broad next-generation methylation panels, (Shaukat et al., 2025; Chung et al., 2024) introduced methodological heterogeneity and may partly explain the variability in sensitivity across studies. Indeed, the sensitivity analyses restricted to next-generation methylation platforms, in which pooled CRC sensitivity increased, and between-study heterogeneity was no longer observed, suggested that technological evolution in assay design represents a key driver of variability in diagnostic performance. Although all included trials enrolled asymptomatic participants, subtle differences in population demographics, sample processing, and analytic thresholds could have influenced performance metrics. In addition, the interval between blood sampling and colonoscopy (up to 120 days in some COVID-affected settings) may represent a potential source of flow and timing bias, although rapid biological progression within this timeframe is unlikely in the context of colorectal carcinogenesis. Finally, the lack of data on cost-effectiveness, interval cancer rates, and longitudinal test reproducibility highlights the need for large-scale, real-world implementation studies to establish how ctDNA testing can be optimally integrated within established CRC screening frameworks.

Further technological advancements are expected to improve the analytical sensitivity of ctDNA assays through ultra-deep sequencing, methylation enrichment, and integration with complementary biomarkers such as proteomic or metabolomic signatures. Progress in computational modelling and risk stratification could enable adaptive, personalised screening intervals. However, real-world validation across diverse populations and healthcare systems remains crucial to ensure generalisability, cost-effectiveness, and equitable access before ctDNA-based screening can be implemented in clinical practice. (Ma et al., 2024; Ladabaum et al., 2020; Tsui et al., 2025)

5. Critical view and scope of the review

This review was restricted to large, prospective, population-based studies conducted in asymptomatic, average-risk individuals, to reflect performance in real-world screening settings rather than enriched diagnostic cohorts. Within this framework, ctDNA testing appears to remain at an early developmental stage for population screening, particularly given its limited sensitivity for APL. However, the improved diagnostic performance observed with newer-generation methylation platforms suggests that technological evolution may enhance consistency and clinical applicability. Further well-designed population-based studies are therefore required to clarify whether next-generation assays can achieve a balance between cancer detection, precancerous lesion identification, and acceptable colonoscopy burden suitable for organised screening programmes.

6. Conclusion

This meta-analysis provides a comprehensive and up-to-date synthesis of the diagnostic accuracy of blood-based ctDNA assays for CRC screening in asymptomatic, average-risk individuals. The findings confirm high specificity and moderate sensitivity for invasive CRC, but consistently low sensitivity for APL, aligning with evidence from recent population-based trials. These results reinforce existing knowledge by outlining the biological and technological limitations of ctDNA testing and its potential as a supplementary, patient-friendly tool to enhance screening participation. Clinically, ctDNA testing may represent an important step towards more accessible, precision-oriented CRC screening strategies, pending further validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.critrevonc.2026.105289.

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